



MATHEMATICS METHODS : UNITS 1 & 2, 2023

PM

Test 1 – (10%)

1.3.1. to 1.3.7, 1.2.19 to 1.2.21, 1.2.24

1.2.1 to 1.2.8, 1.2.13

Time Allowed 28 minutes	First Name	Surname	Marks
			28 marks

Circle your Teacher's Name:

Mrs Alvaro	Mr Bartoshefski	Mr Buckland	Mr Liu
Mr Lockyer	Mr Luzuk	Mr Mohammadi	
Ms Narendranathan	Mr Ng		
Mr Stephens	Mr Tanday		

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1 [4 marks – 2,2]

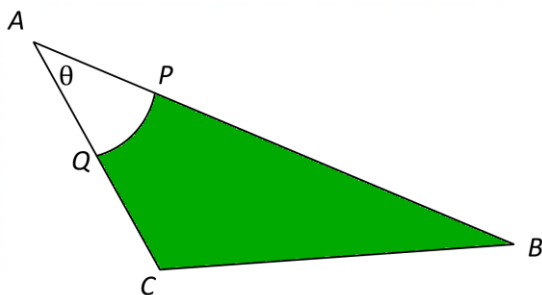
If $f(x) = 2x + 5$ and $g(x) = x^2 - 1$, find;

(a) $f(x^2) - g(x)$

(b) k , such that $f(-2k) = g(0)$

Question 2 [4 marks – 2,2]

Triangle ABC has $AB = 8\text{cm}$ and $AC = 5\text{cm}$. A circle, centre A and radius 3cm intersects AB and AC at P and Q respectively. The area of triangle ABC is $10\sqrt{3}\text{ cm}^2$



Determine the **exact value** of;

- (a) the acute angle θ

- (b) the **area** of the shaded region.

Question 3 [8 marks – 1, 3, 4]

(a) Express 18° in radians.

(b) Evaluate $\sqrt{3}\sin\left(\frac{2\pi}{3}\right) - \tan^2\left(\frac{5\pi}{6}\right)$

(c) Points $B(a,b)$ and $D(c,d)$ are plotted on the unit circle, where

$$\angle COB = 32^\circ \text{ and } \angle DOE = \theta^\circ$$

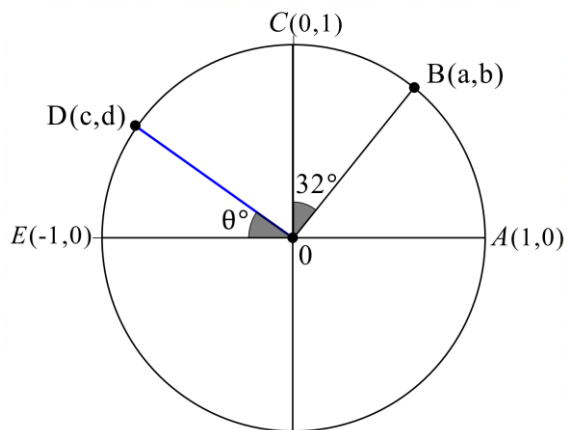
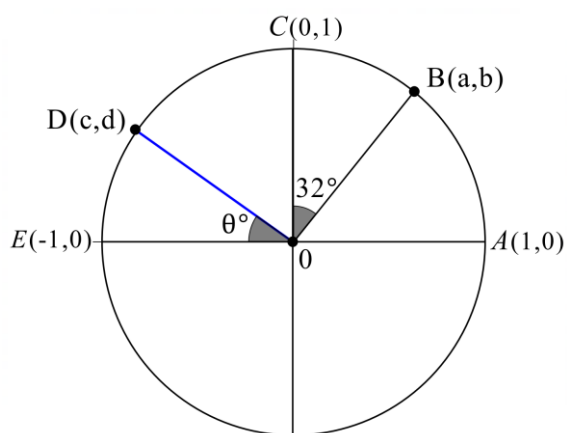
In terms of a, b, c and/or d ,
find ;

(i) $\sin 58^\circ$

(ii) $\cos \theta$

(iii) $\tan 122^\circ$

(iii) $\cos(\theta + 90^\circ)$



Question 4 [4 marks]

Complete the following table. Where no domain is stipulated, assume the natural domain

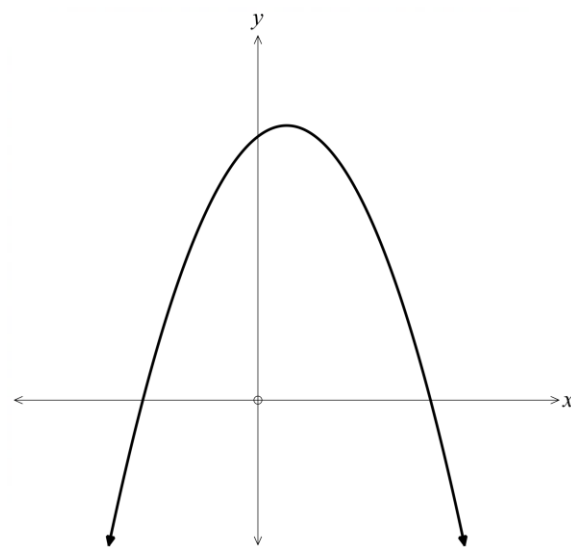
Function	$y = 3 - x$	$y = -(x - 3)^2 - 4$	$y = \frac{1}{x + 5} + 3$	$y = \sqrt{x - 8}$
Domain	$2 < x \leq 7$			
Range				

Question 5 [3 marks]

Find the equation of the straight line through the point $(3, 5)$, which is perpendicular to the straight line passing through the points $(4, -6)$ and $(2, 3)$.

Question 6 [5 marks- 2, 3]

The graph of $y = f(x) = -x^2 + x + k$ is drawn below.
The discriminant is equal to 25, ($\Delta = 25$).



- (a) Show that the value of $k = 6$.
- (b) Hence or otherwise, state the equation of the graph in the form $y = b(x - c)^2 + d$

Extra working page:



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PM

Test 1 – (10%)

1.3.1. to 1.3.7, 1.2.19 to 1.2.21, 1.2.24
1.2.1 to 1.2.8, 1.2.13

Time Allowed 21 minutes	First Name	Surname	Marks
			21 marks

Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu
Mr Lockyer Mr Luzuk Mr Mohammadi
Ms Narendranathan Mr Ng
Mr Stephens Mr Tanday

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART B – CALCULATOR ASSUMED

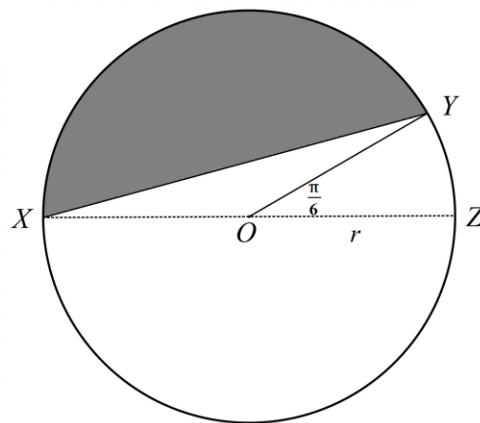
All approximate answers to be stated to an accuracy of 2 decimal places, unless otherwise directed

Question 7 [3 marks]

A circle with centre O , and radius r ,

has $\angle YOZ = \frac{\pi}{6}$

Determine the area of the shaded region if the radius is 10 cm .



Question 8 [7 marks – 4,3]

A yacht is located at point Y and is sailing on a bearing $032^\circ T$ towards a lighthouse at point L , 1.5 km from point Y .

From point Y , the yacht's navigator spots a boat at point B bearing $120^\circ T$.

The bearing of the lighthouse from the boat at B is $332^\circ T$.

(a) **Draw a diagram** and calculate the distance between the yacht and the boat.

(b) From the boat, the top of the lighthouse has an elevation of 2° . Determine the height (**to the nearest metre**) of the lighthouse above sea level.

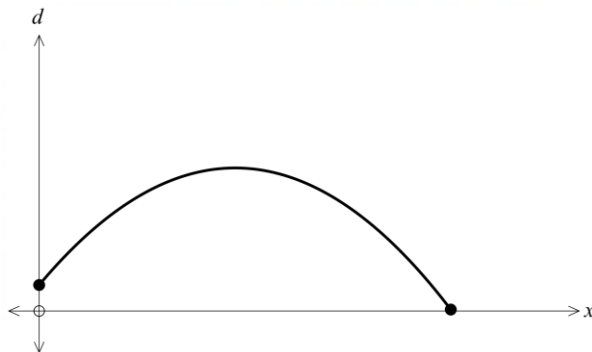
Question 9 [3 marks]

For the quadratic function $y = f(x)$, $f(2) = 0$ and $f(-4) = 0$.
If the minimum value is -6 , find the rule for $y = f(x)$.

Question 10 [4 marks-1,1,2]

Ali, the cricketer, struck a ball such that its height, d metres , after it had travelled x metres horizontally, was given by the rule

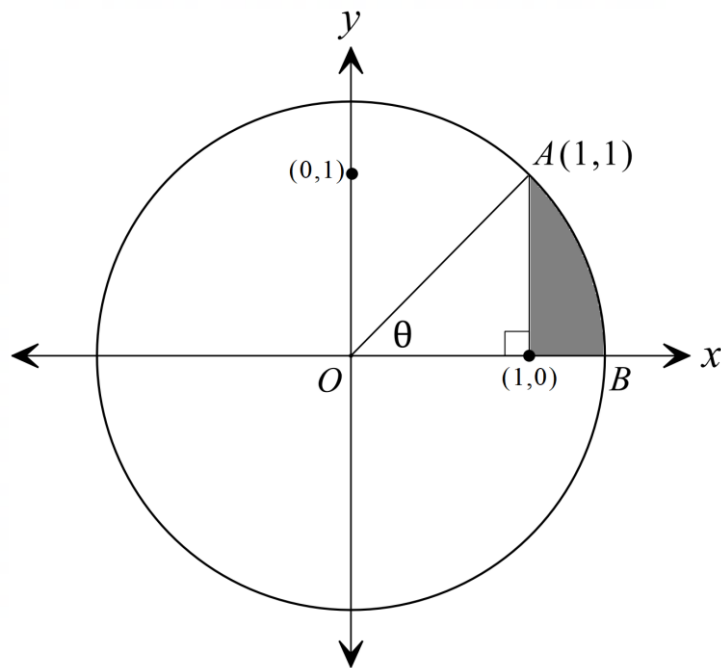
$$d(x) = 1 + \frac{3}{5}x - \frac{1}{50}x^2 , \quad x \geq 0$$



- (a) At what height was the ball when it was struck ?
- (b) What was the maximum height reached by the ball?
- (c) If the fielder, Sunny, who was more than 10m away from batter, caught the ball when it was 2 m above ground, how far was Sunny from the batter ?

Question 11 [4 marks – 1, 2, 1]

Consider the circle drawn below, centred at O , the origin $(0,0)$. The line segment connecting O , and the point $A(1,1)$ makes an angle of θ with the positive x -axis



- (a) Determine the coordinates of point B
- (b) Determine that the **exact length** of the minor arc AB
- (c) Hence, or otherwise, find the **exact perimeter** of the shaded region

Extra working page: